

INVITED TALK / ABSTRACT

Singularity models for the one-phase free boundary problem

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Abstract

Free boundary problems are problems of solving an elliptic or parabolic PDE, but on a domain that is determined by the solution, and they arise naturally in the study of water waves, the propagation of flames, etc. In this talk, I will focus on the regularity of the one-phase free boundary problem (also called the Bernoulli problem). In particular, I will highlight some of its connection with the regularity theory of minimal surfaces, including new examples of singularity models from isoparametric hypersurfaces, local and global asymptotics of solutions to the problem near a homogeneous solution with isolated singularity.

Keywords: one-phase free boundary problem; singularity models; elliptic PDE; parabolic PDE; Bernoulli problem; regularity; minimal surfaces; isoparametric hypersurfaces; isolated singularity